# Soham Himanshubhai Padia

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## EDUCATION

**Northeastern University, Boston, MA** **Sep 2025 – May 2027**
Master of Science in Artificial Intelligence at Khoury College of Computer Sciences GPA: **4.0/4.0**
*Graduate Teaching Assistant:* CS5800 Algorithms (Jan 2026 – Present; reappointed for Summer 2026).
*Self-study:* ARENA – Ch 0 Fundamentals, Ch 1 Transformer Interpretability (core exercises completed).

**Dwarkadas J. Sanghvi College of Engineering, Mumbai, India** **Jun 2021 – May 2025**
Bachelor of Technology in Information Technology GPA: **3.4/4.0**

## RESEARCH EXPERIENCE

**Pre-Capitulation Detection in Multi-Turn LLM Sycophancy** | Advised by Prof. Malihe Alikhani, NEU **Jan 2026 – May 2026**

- Co-first authored an **EMNLP 2026 submission** investigating whether sycophantic capitulation emerges **internally before behavioral flips** in multi-turn LLM interactions under monotonically escalating social pressure.
- Built a multi-turn hidden-state extraction and probing pipeline across **5 open-weight LLMs** (Llama-3.1-8B, Qwen2.5-7B, Qwen3.5-9B, DeepSeek-R1-7B, Gemma-2-9B), collecting layer-wise activations over **474 conversations** and training probes on first-flip-truncated labels validated by LLM-as-judge.
- Found that **Turn 0 to Turn 1 cosine similarity** predicts eventual capitulation above chance (peak AUC **0.60–0.64**) with no classifier training, while linear probes fail at every layer and nonlinear classifiers exceed them by **6.6 pp** on average – evidence that pre-flip signals are nonlinearly encoded.

**Mechanistic Direction Extraction & Evaluation-Aware Behavior** | Informal mentorship from Bau Lab **May 2026 – Present**

- Investigating whether negative-projection sequences are semantically opposite or merely **geometrically opposite** – early evidence suggests strong negative-direction triggers are artifact-like tokens (code fragments, punctuation) rather than coherent anti-social language.
- Studying **evaluation-aware behavior** via indirect measurement of LLM internal states induced by emotional prompts (extending Spiller et al., npj Digital Medicine 2025), designing behavioral and hidden-state proxies that avoid tipping the model off to the assessment.

**Spectral Interpretability of Attention QK Matrices** | Independent research **May 2026 – Present**

- Designing experiments to decompose $W_{QK} = W_{\mathbf{sym}} + W_{\mathbf{skew}}$ per head in pretrained transformers (GPT-2 Small, Pythia-160M, BERT-base) and eigendecompose $W_{\text{sym}}$ to systematize Anthropic's positive-eigenvalue induction-head statistic across all heads.
- Planning correlated study of $W_{\text{skew}}$ structure in decoder middle layers (extending Saponati et al., ICML 2025), with explicit RoPE deconfounding and magnitude-matched ablations to separate symmetry effects from norm effects.

## PUBLICATIONS

- **Before the Model Caves: Detecting Pre-Capitulation States in Multi-Turn Sycophancy.** S. Padia, U. Saraogi, T. D'Avola, V. Shah. Under submission, EMNLP 2026.
- **Adaptive Trust Consensus for Blockchain IoT.** S. Padia et al. arXiv:2512.22860 (2025). [link]
- **NLP Techniques in Chi-Square Tests and Feature Selection.** Springer (2025). [doi]

## SELECTED PROJECTS

**Transformer Mechanistic Interpretability** | Python, PyTorch, TransformerLens

- Performed activation patching across **15+ contrastive GPT-2 input pairs** to localize causally relevant attention heads influencing sentiment behavior.
- Executed targeted head ablations; removing the top **5 causal heads** shifted the sentiment metric by **+0.252**, with fixed-seed experiments and documented failure cases.

**Staged Attention Architecture Exploration** | PyTorch, custom training loop **Apr 2026**

- Designed and implemented a scout-refiner attention framework with a custom spec language allowing serial within-layer head composition; ran 10-variant ablation against parallel baseline.
- **Negative result:** serial execution caused 2.36× wall-clock slowdown (memory-bandwidth pipeline bubbles) with negligible quality gain at matched compute; results traced to Elhage/Olsson cross-layer composition argument.

## TECHNICAL SKILLS

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| **ML/AI** | PyTorch, TransformerLens, NNsight/NDIF, HuggingFace, deep learning, NLP, RL/MARL, interpretability, probing |
| **LLM/GenAI** | Hidden-state extraction, activation patching, direction extraction, LLM-as-judge evals, RAG, prompt engineering |
| **Systems** | SLURM/HPC, Docker, FastAPI, CI/CD, Prometheus, Linux, Git, AWS, distributed inference |
| **Evaluation** | Probe training, cross-validation, Fisher analysis, inter-judge reliability, reproducibility, checkpointing |